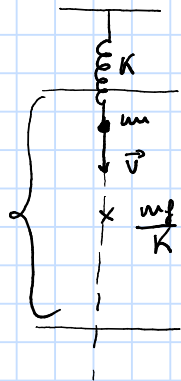
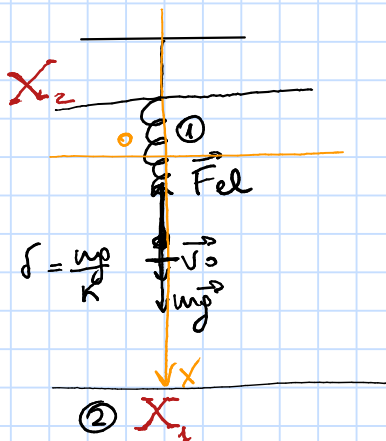


1)



quale sarà l'ampiezza dell'oscillazione?
 Quale il massimo allungamento della molla?
 Quale la massima compressione?



Forze conservative $\rightarrow$ vale la legge di conservazione di E_m

$$\Delta E_m = 0$$

$$E_p(2) = \frac{Kx^2}{2} - mgx \quad E_p(1) = 0$$

$$E_m(1) = 0 - \frac{1}{2}mv_0^2 \quad E_c(1) = -\frac{1}{2}mv_0^2$$

$$E_m(1) = -\frac{1}{2}mv_0^2$$

$$E_m(2) = \frac{Kx^2}{2} - mgx + 0 \quad E_c(2) = 0$$

con x massima estensione della molla X_1

$$E_m(2) = \frac{KX^2}{2} - mgX$$

$$\Delta E_m = 0 = E_m(2) - E_m(1) = 0$$

$$\frac{KX^2}{2} - mgX + \frac{1}{2}mv_0^2 = 0$$

$$\frac{mg \pm \sqrt{mg^2 - Km v_0^2}}{K} = \frac{mg \pm \sqrt{mg^2 \left(1 - \frac{K v_0^2}{mg^2}\right)}}{K} = \frac{mg \pm mg \sqrt{1 - \frac{K v_0^2}{mg^2}}}{K} = \frac{g \pm g \sqrt{1 - \frac{K v_0^2}{mg^2}}}{K} = X_{1,2}$$